

Core Java

Course Duration: 30Hrs

S. No	Content
1	Introduction of Java • What is Java • Who Developed Java • How Java runs • Features of Java • Applications and Types of Java
2	Fundamentals of Java • Main function in java • Hello World Program • Variables • Naming rule for variables • Variable Declaration
3.	Datatypes What is Datatypes Declaring a datatype Types: 1. Primitive 2. Non- primitive

4.	Operators • Arithmetic • Assignment • Relational • Increment / Decrement • Logical • Conditional
5.	Decision Making ○ If Statements a. If – else b. Nested If c. If else ladder ○ Switch Case.
6	Flow Control • For Loop • While loop • Do-While loop.
7	Methods • Introduction to methods • How to create a method • Uses of methods • Types of Methods

8	Array • Introduction to array • How to declare a array • Getting user input for array Types: 1. Single Dimensional 2. Multi-Dimensional
9	String • Introduction to String • Declaration of String • String methods • String Pool
10	String Buffer • Introduction to String Buffer • Declaration of String Buffer • String Buffer methods String Builder • Introduction to String Builder • Declaration of String Builder • String Builder methods • Difference between String, String Buffer & Builder

11	Object Oriented Programming - OOPs • Introduction of OOPs • Concepts or types of OOPs • Access Modifier • Class & Objects
12	Inheritance Introduction to inheritance Using extends keyword Types: 1. Single 2. Multi-level 3. Hierarchy
13	Abstraction • Introduction to abstraction • Using abstract keyword • Declaration of abstract method • Defining abstract method
14	Interface • Introduction to interface • Creating an interface using interface keyword • Declaring a variable and method • Multiple - inheritance

15	Encapsulation • Introduction to Encapsulation • Using private keyword • Setter & getter Method
16	Polymorphism • Introduction to polymorphism • Method Overloading • Constructor Overloading I. Introduction to constructor II. Creating a constructor III. Types: ◦ Default Constructor ◦ Parameterized Constructor • Method Overriding
17	Exception Handling • What is Exception • How we can handle the exception Keywords: 1. Try Catch 2. Throw 3. Throws 4. finally
18	Exercise & Projects.